Project 2

Developing an AI Application

This project not only guides you through building an AI application but also immerses you in the full product development lifecycle: from ideation to deploying an interactive AI system, and collaborating effectively within a team.

1 Introduction

In this project, students work collaboratively as a team of motivated and creative individuals to conceptualize and develop an innovative AI application. Each group is encouraged to freely select the problem they want to solve and the AI techniques they wish to apply (e.g., classification, recommendation, chatbot, computer vision, etc.). Teams should design a unique and meaningful solution that reflects their technical skills, creativity, and problem-solving mindset.

Keep in mind that this project **does NOT require building AI models from scratch**. Therefore, you are encouraged to use existing frameworks, pretrained models, and APIs such as:

- Hugging Face: https://huggingface.co/
- Google Cloud AI: https://cloud.google.com/ai
- OpenAI API: https://platform.openai.com/
- TensorFlow Hub: https://tfhub.dev/
- PyTorch Hub: https://pytorch.org/hub/
- etc.

In addition, here are some suggested tools to support teamwork:

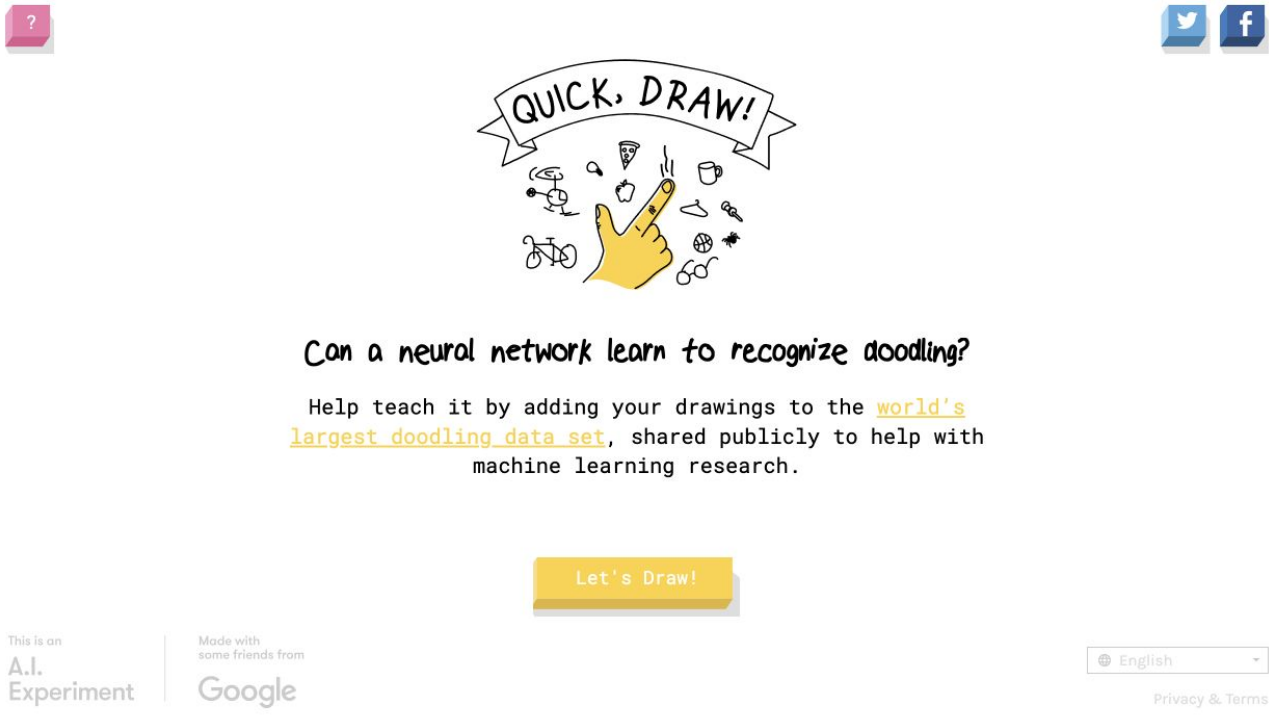
- Zoom: https://zoom.us/
- Slack: https://slack.com/
- Trello: https://trello.com/
- Discord: https://discord.com/
- GitHub: https://github.com/ (for code repository)
- etc.

2 Examples of AI Applications

Explore the applications below to understand what an AI application can look like. Each example illustrates a concrete input and a useful output. Use these examples as inspiration for a focused application that your team can develop within the project duration.

2.1 Image Recognition and Object Detection

- **Quick, Draw! (Google)**: Draw an object on the screen; the application predicts what you are drawing. This illustrates sketch classification in an interactive game.
  Try online | YouTube demo (Google for Developers).



Hình 1: Quick, Draw! homepage, inviting users to try an interactive sketch-recognition game. Source: Google Quick, Draw!.

- **Object Detection Web**: Upload an image; the application displays labels and bounding boxes around detected objects.
  Try online (Hugging Face).

2.2 Image Background Removal

- **Remove Background Web**: Upload a photo; the application separates the main subject from its background and produces an image with a transparent background.
  Try online (Hugging Face).

2.3 Speech Recognition and Speech Synthesis

- **Whisper**: Upload a speech recording; then the application converts the speech into a text transcript. This can inspire an application for transcribing interviews or lecture recordings.
  Try online (Hugging Face).

- **Kokoro Web**: Enter a short text; the application generates spoken audio. This can inspire a tool for reading written content aloud.
  Try online (Hugging Face).

2.4 Text Classification and Translation

- **Zero-Shot Classification Demo**: Enter a text and candidate labels; the application scores how well each label matches the text. For example, try sorting messages into topics such as study, sports, or travel.
  Try online (Hugging Face).

- **DeepL Translator**: Enter a paragraph and choose a target language; the application returns a translation.
  Try online.

2.5 Document Question Answering and Summarization

- **Gemini Notebook (formerly NotebookLM)**: Add a document or webpage and ask a question; the application generates answers or summaries with references to the supplied sources. A focused project could let students ask questions about a course document.
  Try online (Google sign-in required) | YouTube demo (Google Cloud, NotebookLM).

You can explore more applications in the Hugging Face Spaces directory.

These examples range from simple to complex, but by starting small, working together, and improving step by step, you can build an AI application of your own.

3 Required Deliverables

3.1 AI Application

Your AI application must satisfy the following requirements:

- A clear and meaningful problem definition relevant to real-world use cases.
- A system that demonstrates the capabilities of your AI application (prediction, classification, generation, detection, etc.).
- A simple and user-friendly interface (web, desktop, or mobile). You may use tools such as Gradio, Streamlit, Flask, Tkinter, Flutter, etc.

3.2 First Submission - Progress Report

Prepare a brief progress report one week after the project is released, which includes:

- **Project Overview**: Summarize the addressing problem and the proposed AI solution.
- **Current Progress**: Describe completed tasks, each team member's contributions, and any preliminary results or prototype screenshots.
- **Challenges**: Identify difficulties encountered and how your team plans to address them.
- **Next Steps**: List the remaining tasks, assigned team members, and planned timeline for completion.

3.3 Final Submission

Your report and slide must be submitted in a compressed ZIP file (`.zip` extension) named according to the format **TeamName.zip**. If the compressed file is larger than 20MB, you may upload it to Google Drive and provide a text file with the public shared link (absolutely **no modifications** are allowed after the deadline).

Here is a detail of the directory organization:

```
TeamName
├── Report.pdf
├── Slide.pdf
└── GitHub_URL.txt (the instructor can verify your implementation process)
```

3.3.1 Final Report

Prepare a detailed report that includes:

- **Project Planning and Task Distribution**: Document the team member responsibilities, which includes information on each task assigned to team members and the completion rate. E.g., Student A has percentage of completion 90% and the group work has total score of 9.0, then A receives a score of 9.0 * 90% = 8.1.

  Therefore, it is important to evaluate the contribution of group members fairly.
- **Pretrained models or API**: Briefly describe the chosen pretrained model or API.
- **System Design and Architecture**: Describe the components and main functionalities.
- Include any references in a properly formatted bibliography section.

Sinh viên thuộc hệ TCTA vui lòng viết báo cáo bằng tiếng Anh (hệ Chính Quy không bắt buộc).

3.3.2 Slide

- Make clear slides that show the main points of your project.
- Prioritize images and avoid too much text.

3.3.3 Presentation

- All members must participate in the presentation. You may use either English or Vietnamese.
- The presentation should be coherent, impressive, and fit within a 10-minute time limit.

4 Assessment

No.	Details	Score
1	AI Application	5 points
2	Progress Report	1 point
3	Final Report	2 points
	Slide	1 point
	Presentation	1 point
	Total	**10 points**

5 Notices

Please pay attention to the following notices:

- This is a **GROUP** assignment. Each group has 4 - 5 members.

  Groups with fewer or more members than the specified limit require lab instructor approval.

- Duration: approximately 3 weeks.

- AI tools are **not restricted**; however, students should use them wisely. Lab instructors have the right to conduct additional oral interviews with random groups to assess their knowledge of the project.

- Any form of plagiarism, dishonesty, or misconduct will result in a grade of zero for the course.

The end.